

Matroid Duality

Def'n: Given $M = (E, \mathcal{I})$, $M^* = (E, \mathcal{I}^*)$ is
s.t. $I \in \mathcal{I}^*$ iff $E \setminus I$ is a base in $\mathcal{I}$.

Ex: For $G = (V, E)$ a connected graph,
then $I \subseteq E$ is an independent set of M_G^*
iff $E \setminus I$ is a spanning tree.

M^* is a Matroid

Thm: M^* is a matroid with rank fn

$$r^*(S) = |S| - (r(E) - r(E \setminus S)). //$$

pf: $r^*(S \cup T) + r^*(S \cap T) = |S \cup T| + |S \cap T| - 2r(E)$

$$+ r(E \setminus (S \cup T)) + r(E \setminus (S \cap T))$$

$$= |S| + |T| - 2r(E) + r(\underline{E \setminus S} \cap \underline{E \setminus T}) + r(\underline{E \setminus S} \cup \underline{E \setminus T})$$

$$\leq |S| + |T| - 2r(E) + r(E \setminus S) + r(E \setminus T)$$

$$= r^*(S) + r^*(T)$$

Hence: r^* is submodular. (clearly, $r(\emptyset) = 0$ and its marginal values are 0, 1.

$\Rightarrow \tilde{M} = (E, \tilde{\mathcal{I}})$, $I \in \tilde{\mathcal{I}} \Leftrightarrow r^*(I) = |I|$ is a matroid.

I a base of $M^* \Leftrightarrow r^*(I) = |I| = r^*(E) \Leftrightarrow r(E \setminus I) = r(E) \Leftrightarrow |I| = |E| - r(E)$

$\Leftrightarrow E \setminus I$ is a base of M .



Deletion and Contraction

Let $M = (E, \mathcal{I})$ be a matroid

Defn: • Deletion of $e \in E$ produces

$$M \setminus e = (E - e, \{I \subseteq E - e, I \in \mathcal{I}\})$$

• Contraction of $e \in E$ produces

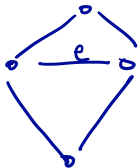
$$M / e = (E - e, \{I \subseteq E - e; I + e \in \mathcal{I}\})$$

Rank: • $r_{M \setminus e}(S) = r_M(S) \quad \forall S \subseteq E - e$

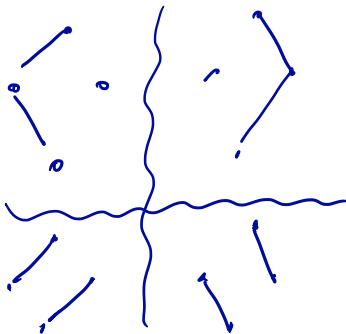
• $r_{M / e}(S) = r_M(S + e) - r_M(e) \quad \forall S \subseteq E - e$

A Small Example

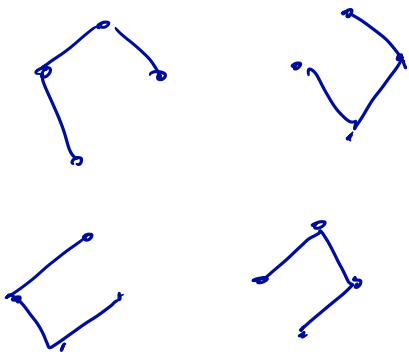
$$G = (V, E)$$



Bases of M/e



Bases M/e



Deletion, Contraction and Duality

Lemma : $(M/e)^* = M^* \setminus e$

pf: Case (i) : $r(e) = 0$

Then, no base of M contains $e \Rightarrow$ every base of M^* contains e .

Then bases of M/e are the same as the bases of M and hence are the comp. of bases of M^* minus e .

Case (ii) : $r(e) = 1$

I ind in $(M/e)^* \Leftrightarrow I \subseteq B$ for B a base

$\Leftrightarrow I \subseteq (E - e) \setminus B'$ for B' a base of $M/e \Leftrightarrow I \subseteq E \setminus B''$

for a base $B'' = B' + e$, a base of $M \Leftrightarrow I \subseteq B'''$ for some

base B''' of M^* , and $e \notin I \Leftrightarrow I$ is independent in $M^* \setminus e$

Summary of Duality

M

contraction
base

ind. set

circuit

loop

M^*

deletion

complement of base

compl. of spanning set

Cut: minimal set that intersects every base

bridge: every base contains

The Spanning Set Polytope

$$P_{\text{spanning set}}(\mathcal{M}) = P_{ss}(\mathcal{M}) = \text{conv} \left\{ x^S : \begin{array}{l} S \text{ a spanning} \\ \text{set of } E \end{array} \right\}$$

$$Q_{ss}(\mathcal{M}) = \{ 0 \leq x \leq 1, x(u) \geq r_{\mathcal{M}}(E) - r_{\mathcal{M}}(E \setminus u), \forall u \}$$

Thm: $P_{ss} = Q_{ss}$

pf: $u \subseteq E$ spanning in $\mathcal{M}$ iff $E \setminus u$ indep. in $\mathcal{M}^*$ (def'n)

Thm: $x \in P_{ss}(\mathcal{M}) \iff \underline{1} - x \in P_{\text{inset}}(\mathcal{M}^*)$

$$\iff 1 - x \geq 0, \text{ and: } |u| - x(u) \leq r_{\mathcal{M}^*}(u), \forall u \subseteq E$$

But $r_{\mathcal{M}^*}(u) = |u| + r_{\mathcal{M}}(E \setminus u) - r_{\mathcal{M}}(E)$

$$\Rightarrow x(u) \geq |u| - r_{\mathcal{M}^*}(u) = |u| - |u| + r_{\mathcal{M}}(E) - r_{\mathcal{M}}(E \setminus u)$$

